Chapter 12

Distributed Database Management Systems

Database Systems:
Design, Implementation, and Management

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In this chapter, you will learn:

- What a distributed database management system (DDBMS) is and what its components are
- How database implementation is affected by different levels of data and process distribution
- How transactions are managed in a distributed database environment
- How database design is affected by the distributed database environment

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The Evolution of Distributed Database Management Systems

- Distributed database management system (DDBMS)
  – Governs storage and processing of logically related data over interconnected computer systems in which both **data and processing** functions are **distributed** among several sites
- Centralized database required that corporate data be stored in a single central site
- Dynamic business environment and centralized database's shortcomings spawned a demand for applications based on data access from different sources at multiple locations (PDAs for example)

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Distributed Processing and Distributed Databases

- **Distributed processing**
  – Database's logical processing is shared among two or more physically independent sites
  – Connected through a network
  – For example, the data input/output (I/O), data selection, and data validation might be performed on one computer, and a report based on that data might be created on another computer
- **Distributed database**
  – Stores logically related database over two or more physically independent sites
  – Database composed of **database fragments**

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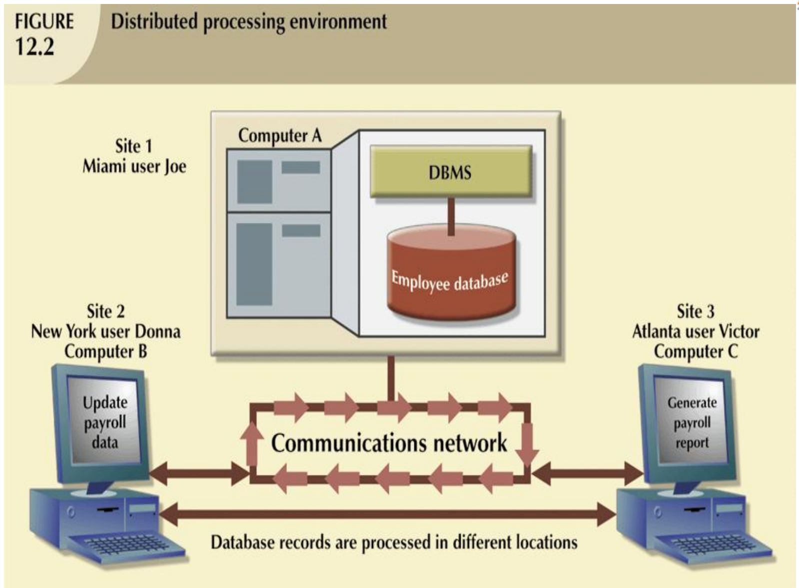

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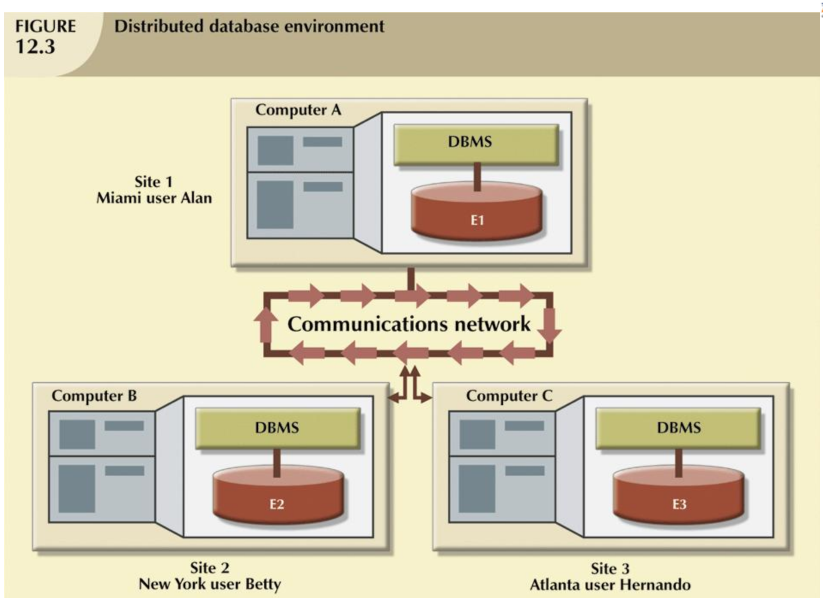

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DDBMS Advantages

- Advantages include:
  - Data are located near "greatest demand" site
  - Faster data access
  - Faster data processing
  - Growth facilitation: New sites can be added to the network without affecting the operations of other sites.
  - Improved communications: Because local sites are smaller and located closer to customers
  - Reduced operating costs: Add workstation not mainframe
  - User-friendly interface
  - Less danger of a single-point failure
  - Processor independence: end user is able to access any available copy of the data, and an end user's request is processed by any processor at the data location.

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DDBMS Disadvantages

- Disadvantages include:
  - Complexity of management and control
  - Security
  - Lack of standards
  - Increased storage requirements: Multiple copies of data are required at different sites
  - Increased training cost

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Characteristics of Distributed Management Systems

- Application interface: interact with the end user, application programs, and other DBMSs
- Validation: to analyze data requests for syntax correctness
- Transformation: to decompose complex requests into atomic data request components
- Query optimization: to find the best access strategy
- Mapping: to determine the data location of local and remote fragments
- I/O interface: to read or write data from or to permanent local storage
- Formatting: to prepare the data for presentation to the end user or to an application program
- Security: to provide data privacy at both local and remote databases
- Backup and recovery: to ensure the availability and recoverability of DB in case of a failure
- DB administration
- Concurrency control: to manage simultaneous data access and to ensure data consistency
- Transaction management: to ensure that the data moves from one consistent state to another

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Characteristics of Distributed Management Systems (continued)

- Must perform all the functions of centralized DBMS
- Must handle all necessary functions imposed by distribution of data and processing
  - Must perform these additional functions transparently to the end user

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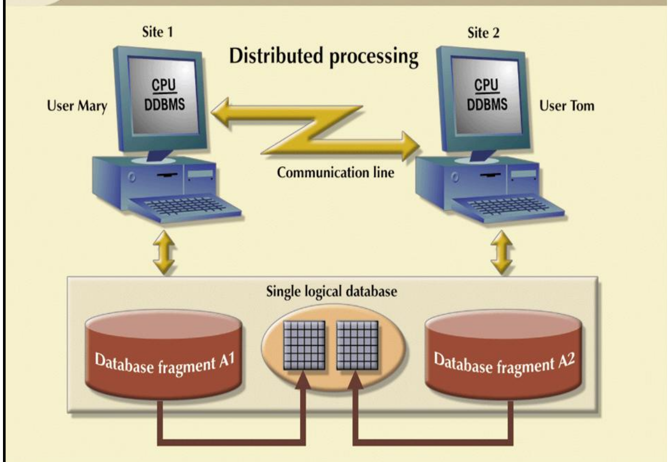
Characteristics of Distributed Management Systems (continued)

FIGURE 12.4 A fully distributed database management system



Both users "see" only one logical database and *do not need to know the names of the fragments*

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Mon 8-7 DDBMS Components

- Must include (at least) the following components:
  - Computer workstations
  - Network hardware and software
  - Communications media
  - Transaction processor (application processor, transaction manager)
    - Software component found in each computer that requests data (***receives and processes*** the application's data requests (remote and local))
  - Data processor or data manager
    - Software component residing on each computer that ***stores and retrieves*** data located at the site
    - May be a centralized DBMS

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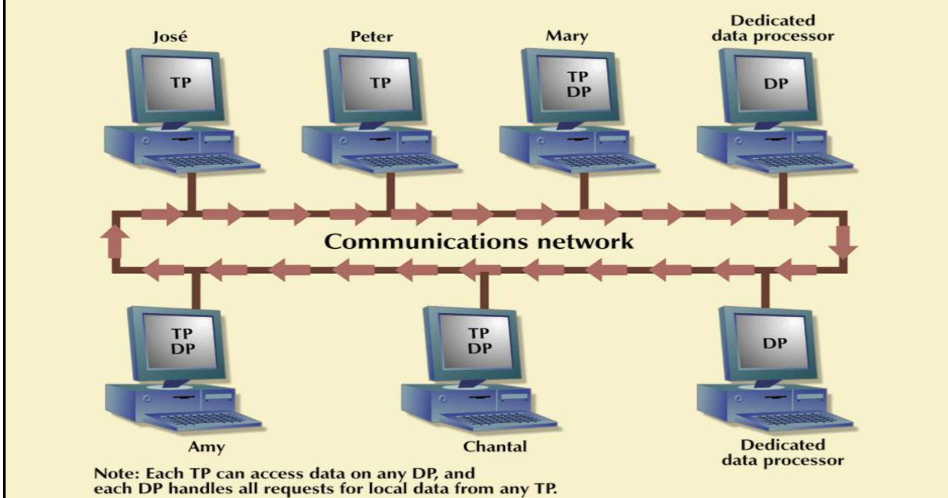
DDBMS Components (continued)

FIGURE 12.5 Distributed database system components



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Levels of Data and Process Distribution

- Current systems classified by how process distribution and data distribution supported

TABLE 12.2 Database Systems: Levels of Data and Process Distribution

	SINGLE-SITE DATA	MULTIPLE-SITE DATA
Single-site process	Host DBMS (mainframes)	Not applicable (Requires multiple processes)
Multiple-site process	File server Client/server DBMS (LAN DBMS)	Fully distributed Client/server DDBMS

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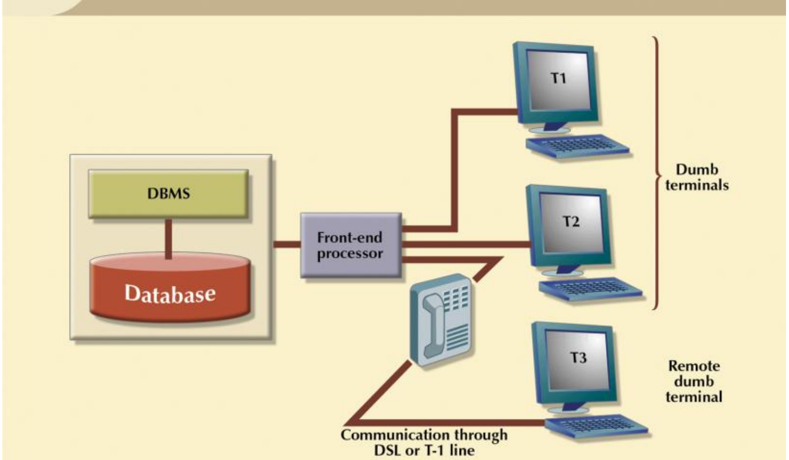
Single-Site Processing, Single-Site Data (SPSD)

- All processing is done on single CPU or host computer (mainframe, midrange, or PC)
- All data are stored on host computer's local disk
- Processing cannot be done on end user's side of system. several processes to run concurrently on a host computer accessing a single DP
- Typical of most mainframe and midrange computer DBMSs
- DBMS is located on host computer, which is accessed by dumb terminals connected to it

FIGURE 12.6 Single-site processing, single-site data (centralized)



Multiple-Site Processing, Single-Site Data (MPSD)

- Multiple processes run on different computers sharing single data repository
- MPSD scenario requires network file server running conventional applications that are accessed through LAN
- Many multiuser accounting applications, running under personal computer network, fit such a description

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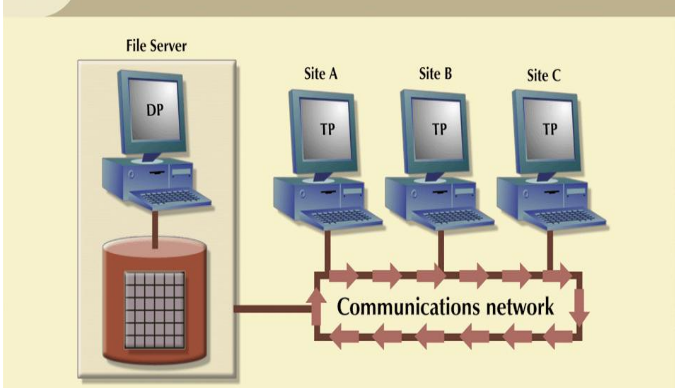
SELECT *
FROM CUSTOMER
WHERE CUS_BALANCE > 1000;
All 10,000 CUSTOMER rows must travel through the network to be evaluated at site A, even if 50 of them have balances greater than $1,000

FIGURE 12.7 Multiple-site processing, single-site data



Client/server architecture is similar to that of the network file server *except that all database processing is done at the server site, thus reducing network traffic*.

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Multiple-Site Processing, Multiple-Site Data (MPMD)

- Fully distributed database management system with support for multiple data processors and transaction processors at multiple sites
- Classified as either homogeneous or heterogeneous
- Homogeneous DDBMSs
  - Integrate only one type of centralized DBMS over a network

Multiple-Site Processing, Multiple-Site Data (MPMD) (continued)

- Heterogeneous DDBMSs
  - Integrate different types of centralized DBMSs over a network
- Fully heterogeneous DDBMS
  - Support different DBMSs that may even support different data models (relational, hierarchical, or network) running under different computer systems, such as mainframes and microcomputers

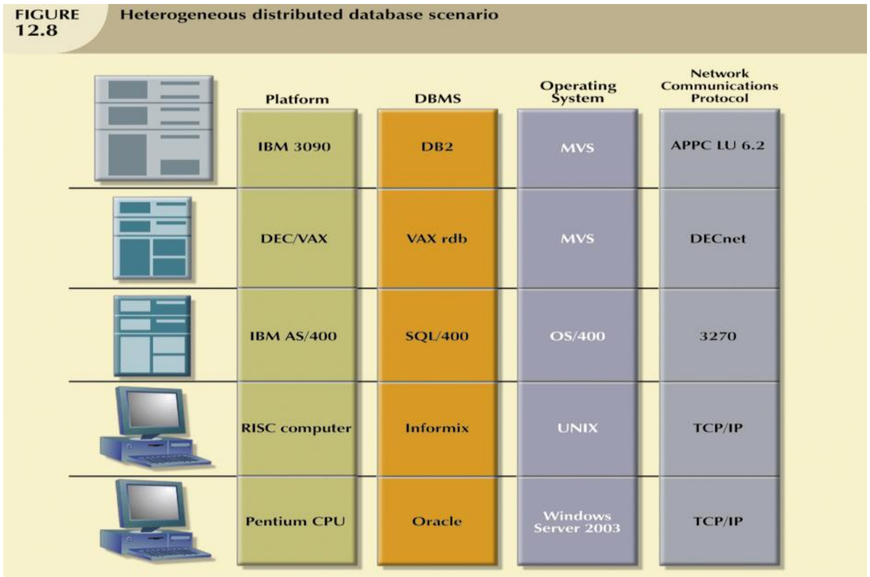

FIGURE 12.8 Heterogeneous distributed database scenario

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Distributed Database Transparency Features

- Allow end user to feel like database's only user
- Features include:
  - Distribution transparency
  - Transaction transparency
  - Failure transparency
  - Performance transparency
  - Heterogeneity transparency

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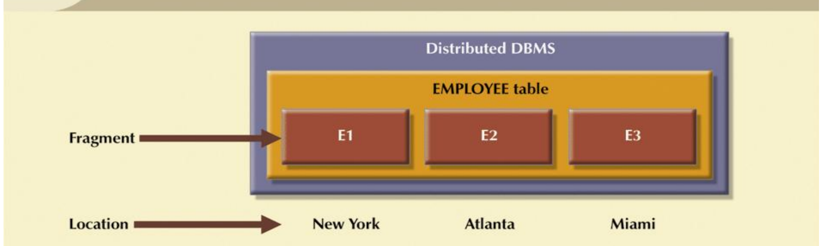
Distribution Transparency

- Allows management of physically dispersed database as though it were a centralized database

FIGURE 12.9 Fragment locations



Transaction Transparency

- Ensures database transactions will maintain distributed database's integrity and consistency
- Ensures transaction completed only when all database sites involved complete their part
- Distributed database systems require complex mechanisms to manage transactions
  - To ensure consistency and integrity

Distributed Requests and Distributed Transactions

- **Remote request:** single SQL statement accesses data from single remote database
- **Remote transaction:** accesses data at single remote site
- **Distributed transaction:** requests data from several different remote sites on network
- **Distributed request:** single SQL statement references data at several DP sites

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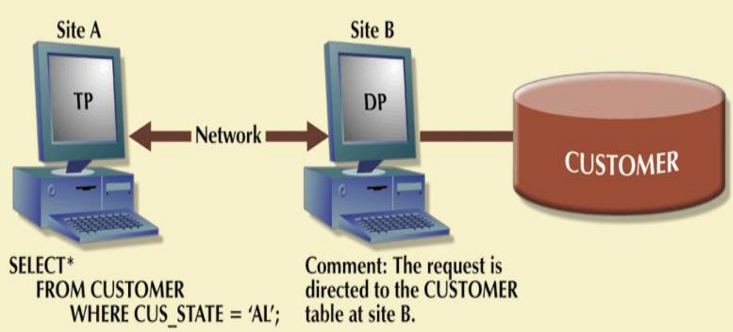
Distributed Requests and Distributed Transactions (continued)

FIGURE 12.10 A remote request



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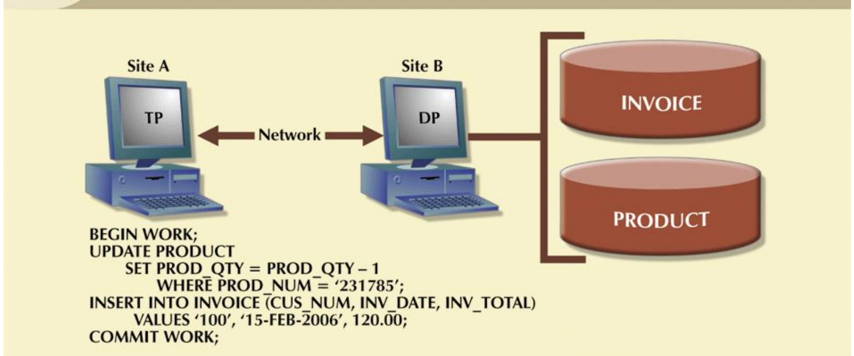
Distributed Requests and Distributed Transactions (continued)

FIGURE 12.11 A remote transaction



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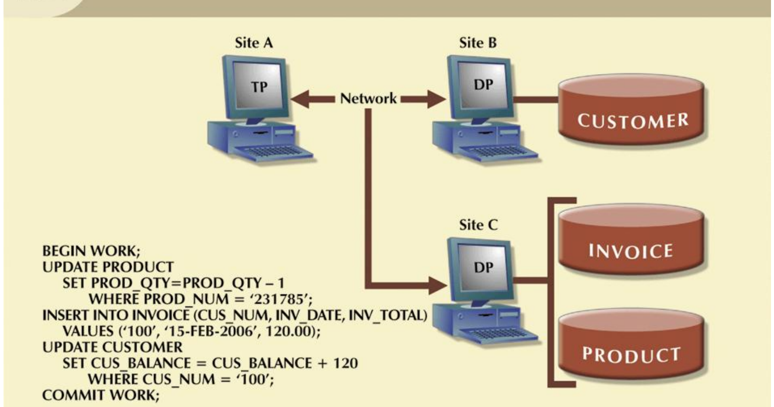
Distributed Requests and Distributed Transactions (continued)

FIGURE 12.12 A distributed transaction



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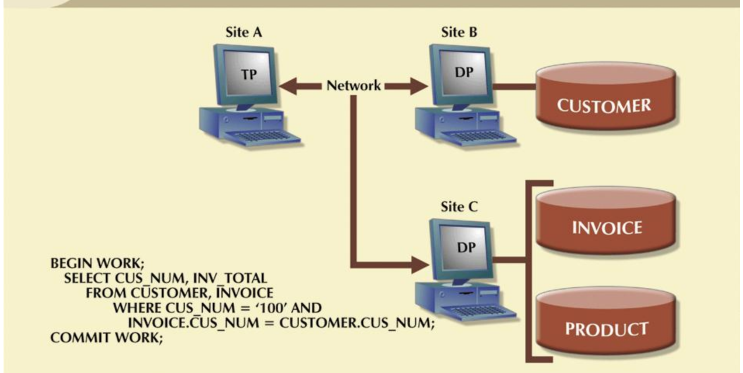
Distributed Requests and Distributed Transactions (continued)

FIGURE 12.13 A distributed request



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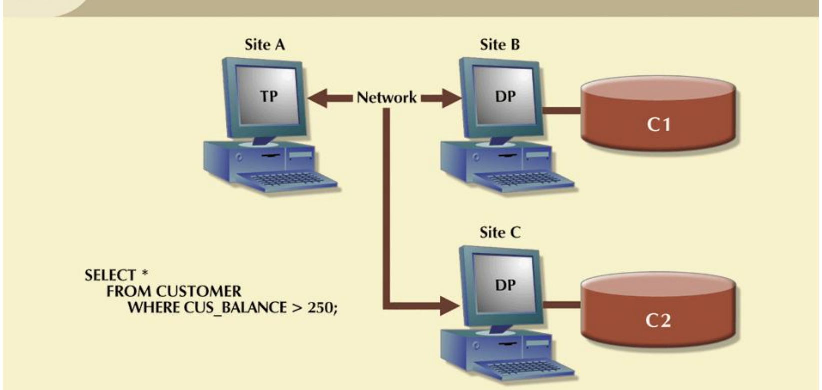
Distributed Requests and Distributed Transactions (continued)

FIGURE 12.14 Another distributed request



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Performance Transparency

- Objective of query optimization routine is to minimize total cost associated with execution of request
- Costs associated with request are function of:
  - Access time (I/O) cost
  - Communication cost
  - CPU time cost
- Must provide:
  - distribution transparency: Allows management of physically dispersed database as though it were a centralized database
  - Replica transparency: DDBMS's ability to hide existence of multiple copies of data from user

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Distributed Database Design

- Design concepts for centralized Database:
  - The Relational Database Model
  - Entity Relationship Modeling; and
  - Normalization of Database Tables
- Three new issues for distributed Database:
  - Data fragmentation
    - How to partition database into fragments
  - Data replication
    - Which fragments to replicate
  - Data allocation
    - Where to locate those fragments and replicas

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Data Fragmentation

- Breaks single object (Db or table) into two or more segments or fragments
- Each fragment can be stored at any site over computer network
- Information about data fragmentation is stored in distributed data catalog (DDC), from which it is accessed by TP to process user requests

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Data Fragmentation (continued)

- Strategies
  - Horizontal fragmentation
    - Division of a relation into subsets (fragments) of tuples (rows)
    - Each fragment represents the equivalent of a SELECT statement, with the WHERE clause on a single attribute.
  - Vertical fragmentation
    - Division of a relation into attribute (column) subsets
    - This is the equivalent of the PROJECT statement in SQL.
  - Mixed fragmentation
    - Combination of horizontal and vertical strategies
    - A table may be divided into several horizontal subsets (rows), each one having a subset of the attributes (columns).

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Data Fragmentation (continued)

FIGURE 12.16 A sample CUSTOMER table

Table name: CUSTOMER

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
10	Sinex, Inc.	12 Main St.	TN	3500.00	2700.00	3	1245.00
11	Martin Corp.	321 Sunset Blvd.	FL	6000.00	1200.00	1	0.00
12	Mynux Corp.	910 Eagle St.	TN	4000.00	3500.00	3	3400.00
13	BTBC, Inc.	Rue du Monde	FL	6000.00	5890.00	3	1090.00
14	Victory, Inc.	123 Maple St.	FL	1200.00	550.00	1	0.00
15	NBCC Corp.	909 High Ave.	GA	2000.00	350.00	2	50.00

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Data Fragmentation (continued)

TABLE 12.4 Horizontal Fragmentation of the CUSTOMER Table by State

FRAGMENT NAME	LOCATION	CONDITION	NODE NAME	CUSTOMER NUMBERS	NUMBER OF ROWS
CUST_H1	Tennessee	CUS_STATE = 'TN'	NAS	10, 12	2
CUST_H2	Georgia	CUS_STATE = 'GA'	ATL	15	1
CUST_H3	Florida	CUS_STATE = 'FL'	TAM	11, 13, 14	3

Company's corporate management requires information about its customers in all three states, but company locations in each state (TN, FL, and GA) require data regarding local customers only.

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Data Fragmentation (continued)

FIGURE 12.17 Table fragments in three locations

Table name: CUST_H1 Location: Tennessee Node: NAS

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
10	Sinex, Inc.	12 Main St.	TN	3500.00	2700.00	3	1245.00
12	Mynux Corp.	910 Eagle St.	TN	4000.00	3500.00	3	3400.00

Table name: CUST_H2 Location: Georgia Node: ATL

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
15	NBCC Corp.	909 High Ave.	GA	2000.00	350.00	2	50.00

Table name: CUST_H3 Location: Florida Node: TAM

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
11	Martin Corp.	321 Sunset Blvd.	FL	6000.00	1200.00	1	0.00
13	BTBC, Inc.	Rue du Monde	FL	6000.00	5890.00	3	1090.00
14	Victory, Inc.	123 Maple St.	FL	1200.00	550.00	1	0.00

Each horizontal fragment may have a different number of rows, but each fragment *must* have the same attributes.

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Data Fragmentation (continued)

TABLE 12.5 Vertical Fragmentation of the CUSTOMER Table

FRAGMENT NAME	LOCATION	NODE NAME	ATTRIBUTE NAMES
CUST_V1	Service Bldg.	SVC	CUS_NUM, CUS_NAME, CUS_ADDRESS, CUS_STATE
CUST_V2	Collection Bldg.	ARC	CUS_NUM, CUS_LIMIT, CUS_BAL, CUS_RATING, CUS_DUE

Suppose the company is divided into two departments: the service department and the collections department. Each department is located in a separate building, and each has an interest in only a few of the CUSTOMER table's attributes.

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Data Fragmentation (continued)

FIGURE 12.18 Vertically fragmented table contents

Table name: CUST_V1 Location: Service Building Node: SVC

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE
10	Sinex, Inc.	12 Main St.	TN
11	Martin Corp.	321 Sunset Blvd.	FL
12	Mynux Corp.	910 Eagle St.	TN
13	BTBC, Inc.	Rue du Monde	FL
14	Victory, Inc.	123 Maple St.	FL
15	NBCC Corp.	909 High Ave.	GA

Table name: CUST_V2 Location: Collection Building Node: ARC

CUS_NUM	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
10	3500.00	2700.00	3	1245.00
11	6000.00	1200.00	1	0.00
12	4000.00	3500.00	3	3400.00
13	6000.00	5890.00	3	1090.00
14	1200.00	550.00	1	0.00
15	2000.00	350.00	2	50.00

Each vertical fragment must have the same number of rows, but the inclusion of the different attributes depends on the key column

Data Fragmentation (continued)

TABLE 12.6 Mixed Fragmentation of the CUSTOMER Table

FRAGMENT NAME	LOCATION	HORIZONTAL CRITERIA	NODE NAME	RESULTING ROWS AT SITE	VERTICAL CRITERIA ATTRIBUTES AT EACH FRAGMENT
CUST_M1	TN-Service	CUS_STATE = 'TN'	NAS-S	10, 12	CUS_NUM, CUS_NAME, CUS_ADDRESS, CUS_STATE
CUST_M2	TN-Collection	CUS_STATE = 'TN'	NAS-C	10, 12	CUS_NUM, CUS_LIMIT, CUS_BAL, CUS_RATING, CUS_DUE
CUST_M3	GA-Service	CUS_STATE = 'GA'	ATL-S	15	CUS_NUM, CUS_NAME, CUS_ADDRESS, CUS_STATE
CUST_M4	GA-Collection	CUS_STATE = 'GA'	ATL-C	15	CUS_NUM, CUS_LIMIT, CUS_BAL, CUS_RATING, CUS_DUE
CUST_M5	FL-Service	CUS_STATE = 'FL'	TAM-S	11, 13, 14	CUS_NUM, CUS_NAME, CUS_ADDRESS, CUS_STATE
CUST_M6	FL-Collection	CUS_STATE = 'FL'	TAM-C	11, 13, 14	CUS_NUM, CUS_LIMIT, CUS_BAL, CUS_RATING, CUS_DUE

Company's structure requires that the CUSTOMER data be fragmented horizontally to accommodate the various company locations; within the locations, the data must be fragmented vertically to accommodate the two departments (service and collection).

Data Fragmentation (continued)

FIGURE 12.19 Table contents after the mixed fragmentation process

Table name: CUST_M1 **Location: TN-Service** **Node: NAS-S**

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE
10	Sinex, Inc.	12 Main St.	TN
12	Mynux Corp.	910 Eagle St.	TN

Table name: CUST_M2 **Location: TN-Collection** **Node: NAS-C**

CUS_NUM	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
10	3500.00	2700.00	3	1245.00
12	4000.00	3500.00	3	3400.00

Table name: CUST_M3 **Location: GA-Service** **Node: ATL-S**

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE
15	NBCC Corp.	909 High Ave.	GA

Table name: CUST_M4 **Location: GA-Collection** **Node: ATL-C**

CUS_NUM	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
15	2000.00	350.00	2	50.00

Table name: CUST_M5 **Location: FL-Service** **Node: TAM-S**

CUS_NUM	CUS_NAME	CUS_ADDRESS	CUS_STATE
11	Martin Corp.	321 Sunset Blvd.	FL
13	BTBC, Inc.	Rue du Monde	FL
14	Victory, Inc.	123 Maple St.	FL

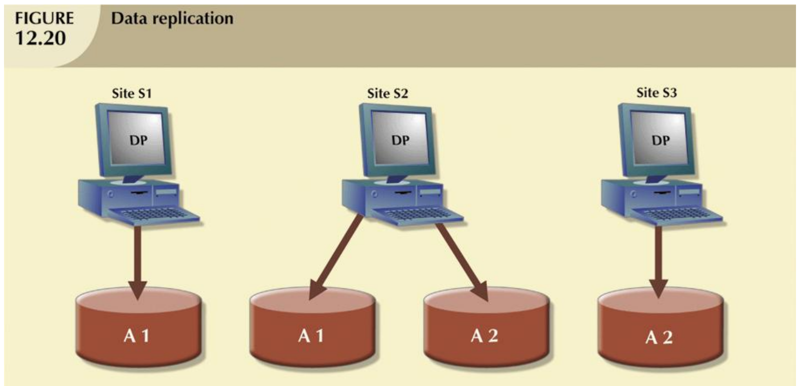
Table name: CUST_M6 **Location: FL-Collection** **Node: TAM-C**

CUS_NUM	CUS_LIMIT	CUS_BAL	CUS_RATING	CUS_DUE
11	6000.00	1200.00	1	0.00
13	6000.00	5890.00	3	1090.00
14	1200.00	550.00	1	0.00

Data Replication

- Storage of data copies at multiple sites served by computer network
- Fragment copies can be stored at several sites to serve specific information requirements
  - Can enhance data availability and response time
  - Can help to reduce communication and total query costs

Data Replication (continued)



Data Replication (continued)

- **Replication scenarios**
  - **Fully replicated database**
    - Stores multiple copies of each database fragment at multiple sites
    - Can be impractical due to amount of overhead
  - **Partially replicated database**
    - Stores multiple copies of some database fragments at multiple sites
    - Most DDBMSs are able to handle the partially replicated database well
  - **Unreplicated database**
    - Stores each database fragment at single site
    - No duplicate database fragments

Data Allocation

- Deciding where to locate data: *which* data to locate *where*
- Data distribution over computer network is achieved through data partition, data replication, or combination of both
- Allocation strategies
  - Centralized data allocation
    - Entire database is stored at one site
  - Partitioned data allocation
    - Database is divided into several disjointed parts (fragments) and stored at several sites
  - Replicated data allocation
    - Copies of one or more database fragments are stored at several sites

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Client/Server vs. DDBMS

- Way in which computers interact to form system
- Features user of resources, or client, and provider of resources, or server
- Can be used to implement a DBMS in which client is the TP and server is the DP

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Client/Server vs. DDBMS (continued)

- **Client/server advantages**
  - Less expensive than alternate minicomputer or mainframe solutions
  - Allow end user to use microcomputer's GUI, thereby improving functionality and simplicity
  - More people in job market have PC skills than mainframe skills
  - PC is well established in workplace
  - Numerous data analysis and query tools exist to facilitate interaction with DBMSs available in PC market
  - Considerable cost advantage to offloading applications development from mainframe to powerful PCs

Client/Server vs. DDBMS (continued)

- **Client/server disadvantages**
  - Creates more complex environment
    - Different platforms (LANs, operating systems, and so on) are often difficult to manage
  - An increase in number of users and processing sites often paves the way for security problems
  - Possible to spread data access to much wider circle of users
    - Increases demand for people with broad knowledge of computers and software
    - Increases burden of training and cost of maintaining the environment